SST Triadic-Casimir / PKLSA Blind Falsifier v0.3.0

Framework-aware, three-source-seal, geometry + regularized filament ODE numerical diagnostic

Run label: REPORT

1. Overall epistemic verdict

Analytical toy controls and a sourced KnotPlot/PKLSA-lineage trefoil geometry were evaluated. The RK4 filament trajectory is numerical simulation only. Physical Euler closure, atom-link coupling, vacuum normalization, metal boundary forces, and gravity are NOT certified.

Official frozen-framework invocation: OFFICIAL_FRAMEWORK_EXECUTED

Framework detail: Canonical framework code ran through real generator, protocol freeze, FULL and reveal-policy check. G2_SPATIAL_FAIL_G3_NOT_RUN.

2. Three independently sealed datasets

D01_EULER SHA-256 seal: e79a4e0cfc370eb08a05d0288e8138a8fcc1b4d9447c7ee80da0764c669a25f4

D02_CLOCK_LINK SHA-256 seal: 60729d7f5a206fdc025dbbde95d9284c54339990e85ba1eaf66dc4331d8ee7eb

D03_TWO_SURFACE SHA-256 seal: 7d5e24b065ee9d6cbd43e3a9ce9a6d0823cf9cee5a6c8b821645e83d9d2991c9

3. Numeric control gates

D01_G01_EULER_DIV: PASS | observed=0.0 | criterion=1e-12

D01_G02_POISSON: PASS | observed=0.0 | criterion=1e-12

D01_G03_POTENTIAL_NULL: PASS | observed=0 | criterion=1e-12

D01_G04_IRROTATIONAL_CLOCK_VARIATION: PASS | observed=0.0008849583697424501 | criterion=1e-05

D01_G07_FILAMENT_RK4_DT_HALVING: PASS | observed=2.0246251170010892e-11 | criterion=0.001

D01_G08_FILAMENT_SPATIAL_REFINEMENT: FAIL | observed=0.08728964728469735 | criterion=0.05

D01_G05_PKLSA_SOURCE_INTEGRITY: PASS | observed=300 | criterion=300

D01_G06_PKLSA_FINITE_BIOTSAVART: PASS | observed=0.002346617627425827 | criterion=finite

D02_G01_GAUGE: PASS | observed=5.551115123125783e-17 | criterion=1e-12

D02_G02_PHASE_RATE: PASS | observed=5.551115123125783e-17 | criterion=1e-12

D02_G03_HOLONOMY: PASS | observed=2.0816681711721685e-17 | criterion=1e-12

D02_G04_RESPONSE_POSITIVE: PASS | observed=0.0 | criterion=0.0

D02_G05_SHEET_STABLE: PASS | observed=1.6305756807341665 | criterion=0.0

D02_G06_NO_RESPONSE_CONTROL: PASS | observed=0.0 | criterion=1e-14

D03_G01_ENERGY_STRESS: PASS | observed=1.141916836020315e-09 | criterion=0.015

D03_G02_DECOUPLED_NULL: PASS | observed=0.0 | criterion=1e-13

D03_G03_FINITE_TOY_RESPONSE: PASS | observed={'negative': 28, 'total': 28} | criterion=all toy coupled configurations

Pilot synthetic/numerical checks: 16/17 PASS

4. PKLSA geometry and numerical bridge

Sourced 3_1 trefoil: KnotPlot relaxed N=300; Fremlin second family registered via E010 but its raw bytes are absent in this bundle.

Source Drive id: 1iOeQe6-jHcBk_NYNwZe3e-N2DQPwohmI

Raw geometry SHA-256: a60b1a95a00110e3cb32087e7b4328d16e967379ebf97d60117b0ec05692ffb2

Points=300; length=16.376343183; segment relative spread=0.00069244

Regularized midpoint Biot-Savart: Gamma=1, eps=0.12, max 150 vs 300 refinement error=0.002346618

The pressure-Poisson source at sample points is computed from numerical strain and curl of the regularized geometric velocity field, with no wall/stress certification.

4b. RK4 finite-core filament simulation

RK4 75 points, fixed T=0.04 and predeclared dt halving, not full Euler and no material boundaries.

5. Model equations

Euler: div(u)=0; omega=curl(u); Laplace(p)=rho*(|omega|^2/2 - ||S||^2).

Conditional clock: N^2=1-|u_eff|^2/c_star^2. PKLSA geometry does not independently determine u_eff for a clock.

Link clock phase: Theta_AB=theta_B-theta_A-sum(path theta_e), conditional on gauge-covariant atom coupling.

One-body response: chi(i xi)=bT (K+xi^2 M)^(-1)b, presently a toy oscillator, not measured metal.

Two-body finite-T toy: F/A=(T/4pi) integral k log(1-r1*r2*exp(-2*k*a)) dk.

RK4 RMS shift=0.013837466; relative dt mismatch=2.02e-11

Fremlin 3_1 raw upstream status: PASS_RAW_HASH_AND_XYZ_PARSER_ONLY

6. Physical qualification blockers

Missing: second authenticated PKLSA bytes, full time-dependent Euler PDE qualification, Kelvin/Floquet modes, finite-core constitutive law, laboratory wall boundary conditions, atom-link coupling, independent reflection operator, fluctuation normalization and independent physical stress.

Never interpret fitted distance exponent or numeric toy attraction as a predicted electromagnetic Casimir force.

7. Reproduction and PDF availability

On Windows: run_all.cmd ALL; run_all.cmd FRAMEWORK; run_all.cmd PKLSA; run_all.cmd VERIFY. A current dynamic PDF is created even if LaTeX/ReportLab are absent.

A CPU framework profile is requested but native physical certification is still blocked by G3. The official canonical SST framework v1.0.7 must exist at the Workbench template location, or SST_FALSIFIER_FRAMEWORK_ROOT must point to it. Its source remains unmodified.

All three v0.3.0 dataset seals are new commitments; the comparator is stored separately for true external-blind review. The earlier v0.1.0 seal is NOT reused as if it still covered the same data.

8. Provenance

Canon v0.8.36: sections 3.5, 3.10, 7.1, 11.16-11.17, S.2, S.52.

A046 v0.3.0 and v0.3.0: Euler / Poisson controls; geometry is not gravity.

Original KnotPlot Drive id and full source sha256 are included in D01. All fields are dimensionless; no prescribed SST rho_f or Casimir amplitude.